

433 Ana Costa Avenue
Suite 81 - Santos/Brazil
11060-003

Telephone: +55 13 33940655
email: surveyors@naabsa.com
www.naabsa.com

Survey Report

Ref:302/2025

Bunker Quantity Survey **MV BENEFACOR** at Santos Port



PERSON / COMPANIES CONTACTED

Costumer	LANGLOIS ENTREPRISES LTD Mr. Stamatis Hahalis
Undersigned Surveyor	NAABSA Marine Surveyors Mr. Wagner de Abreu
Suppliers	Petrobras – Barge SC 60 Ms. Fernanda Amaral Brecale
Vessel's Command	Master / Chief Engineer Mr. Malabosa Aldwin Duerme / Mabao Floro Pedrosa

Contents

1. Background.....	3
2. Vessel's details (Ship's Particulars).....	3
3. Bunker Quantity Survey.....	3
4. Opening Soundings.....	3
4.1. Vessel's tanks.....	3
4.2. Gross Volume (m ³).....	4
4.3. Temperatures.....	4
4.4. Specific gravities.....	4
4.5. Barge's tanks.....	4
4.6. Gross Volume (m ³).....	5
4.7. Temperatures.....	5
4.8. Specific gravities.....	5
5. Closing soundings.....	5
5.1. Vessel's tanks.....	5
5.2. Barge's tanks.....	5
6. Final figures.....	6
7. Photographic Report.....	6
7.1. Vessel's tanks.....	6
7.2. Barge's tanks.....	7
7.3. Sampling.....	8
8. Attachement.....	9

1. Background

The **MV Benefactor** called to **Santos Port** to load a cargo of **corn** in bulk bound to **B.I.K., IRAN**.

Complying with the appointment service from Messrs. **LANGLOIS ENTERPRISES LTD**, we did attend to carry out Bunker Quantity Survey to ascertain the total quantity of fuel delivered.

2. Vessel's details (Ship's Particulars)

Flag : **Liberia**
Port registry : **Monrovia**
Call sign : **5LCR4**
IMO number : **9935466**
Type : **Bulk Carrier**
Delivered : **2022**
LOA : **229,04** m

3. Bunker Quantity Survey

The undersigned surveyor boarded the vessel on **December 12th at 22:00h** l/t, prior to the arrival of the bunker barge.

We introduced ourselves as Bunker Surveyors and proceeded to the ship's office for an initial meeting. During this meeting, we ensured that all safety checklists had been duly verified by the Chief Engineer and the Barge Inspector.

4. Opening soundings

4.1. Vessel's tanks

The opening soundings of the vessel's tanks were carried out by the **Fourth Engineer** for each tank on **December 12th** from **22:20** to **22:40h** local time. The measurements were taken in accordance with standard procedures to ensure accurate assessment of the pre-bunker fuel quantities in the vessel's tanks.

Draft Fore:	5,15	m	Draft Aft:	7,30	m	List:	0,00	Trim Correction Applied:	Yes	METRIC TONS		
Tank n°	Grade	Sounding/ Ullage (m)	Deg °C	Total Vol Observed (m³)	Free Water		Gross Obs Vol (m3)	VCF TAB 54B	Density (S.G.) at 15° C	Gross Std Vol @15° C	IN VAC	IN AIR
					Dip	Vol						
HFO 1P	VLSFO	1,38 u	26,0	284,790	Nil	0,000	284,790	0,9920	0,9470	282,512	267,539	267,228
HFO 1S	VLSFO	4,58 u	26,0	4,200	Nil	0,000	4,200	0,9920	0,9470	4,166	3,946	3,941
HFO 2P	VLSFO	0,11 s	43,0	1,110	Nil	0,000	1,110	0,9797	0,9470	1,087	1,030	1,029
HFO 2S	VLSFO	0,14 s	27,0	1,390	Nil	0,000	1,390	0,9911	0,9256	1,378	1,275	1,274
HFO 3P	VLSFO	4,67 u	35,0	2,260	Nil	0,000	2,260	0,9851	0,9268	2,226	2,063	2,061
HFO 3S	VLSFO	4,68 u	28,0	2,270	Nil	0,000	2,270	0,9906	0,9470	2,249	2,129	2,127
HFO 1SERV	VLSFO	GAUGE	72,0	19,540	Nil	0,000	19,540	0,9571	0,9256	18,702	17,310	17,290
HFO 1 SETT	VLSFO	GAUGE	72,0	14,400	Nil	0,000	14,400	0,9571	0,9256	13,782	12,757	12,742
HFO 2 SERV	VLSFO	GAUGE	88,0	23,800	Nil	0,000	23,800	0,9449	0,9256	22,489	20,815	20,791
HFO 2SETT	VLSFO	GAUGE	88,0	26,000	Nil	0,000	26,000	0,9449	0,9256	24,567	22,740	22,713
OVERFLOW	VLSFO	4,78 u	21,0	4,780	Nil	0,000	4,780	0,9956	0,9312	4,759	4,432	4,426
TOTAL							384,540			377,917	356,036	355,620

Sound: (U)=Ullage / (S)=Deep Sounding

Logbook (VLSFO): **355.780** mt

NAABSA figures.....: **355.620** mt

Difference (mt).....: **+0,160** mt

4.2. Gross volume - m³

Interpolated according to the vessel's sounding table presented at time of calculations. Trim and Heel correction were applied accordingly.



4.3. Temperature

Storage tanks – Display on ECR.

Services and Settling tanks – Display on ECR.

Overflow tank – Sea Water temp.

Engine Room Temp: 38° C / Sea Water Temp: 21° C

4.4. Specific Gravities

Applied according to the vessel's records for bunkering analysis presented by Chief Engineer at time of survey and the current BDN issued by suppliers.

Temperature °C	Specific Gravity
15	0,9470 / 0,9256 / 0,9312 / 0,9268

4.5. Barge's tanks

The opening soundings of the Barge's tanks were carried out by the barge's operator for each tank on December 12th from 23:35 to 23:55h local time. The measurements were taken in accordance with standard procedures to ensure accurate assessment of the pre-bunker fuel quantities in the barge's tanks.

BEFORE TRANSFER							FLOWMETER	N/A m ³				
Draft Fore:	3,40	m	Draft Aft:	3,40	m	List:	0,00	Trim Correction Applied:	YES	METRIC TONS		
Tank nº	Grade	Sounding / Ullage	Deg C	Total Vol Observed	Free Water Dip Vol		Gross Obs Vol (m3)	VCF TAB 548	Density (S.G.) at 15° C	Gross Std Vol @15° C	IN VAC	IN AIR
1 BB	VLSFO	2,902	43,0	224,713	Nil	0,000	224,713	0,9796	0,9452	220,129	208,066	207,824
1 BE	VLSFO	2,895	42,0	226,090	Nil	0,000	226,090	0,9804	0,9452	221,659	209,512	209,268
2 BB	VLSFO	1,698	45,0	405,892	Nil	0,000	405,892	0,9782	0,9452	397,044	375,286	374,849
2 BE	VLSFO	1,703	45,0	402,835	Nil	0,000	402,835	0,9782	0,9452	394,053	372,459	372,026
3 BB	VLSFO	4,426	45,0	164,157	Nil	0,000	164,157	0,9782	0,9452	160,578	151,779	151,602
3 BE	VLSFO	4,441	45,0	162,212	Nil	0,000	162,212	0,9782	0,9452	158,676	149,980	149,806
4 BB	VLSFO	1,728	45,0	403,994	Nil	0,000	403,994	0,9782	0,9452	395,187	373,531	373,096
4 BE	VLSFO	1,696	46,0	406,693	Nil	0,000	406,693	0,9774	0,9452	397,502	375,719	375,281
5 BB	VLSFO	2,864	44,0	254,536	Nil	0,000	254,536	0,9789	0,9452	249,165	235,511	235,237
5 BE	VLSFO	2,969	36,0	245,658	Nil	0,000	245,658	0,9847	0,9452	241,899	228,643	228,377
Total							2396,586	0,9782	0,9452	2344,827	2,216,332	2,677,365

Barge's inspector figures.....: 2,683.388 mt
Surveyor's figures.....: 2,677.365 mt
Difference.....: - 5.993 mt
Flowmeter readings.....: OUT OF ORDER

4.6. Gross volume - m³

Interpolated according to the barge's sounding table presented at time of calculations. The trim correction was applied accordingly.

4.7. Temperature

Storage tanks – Glass thermometer.

4.8. Specific Gravities

According to the BDN issued by suppliers.

Temperature °C	Specific Gravity
15	0,9452
20	0,9419

5. Closing soundings

5.1. Vessel's tanks

The final sounding on the vessel's tanks took place immediately after the Barge's Inspector confirmed that the total amount ordered had been transferred from the barge tanks to the vessel's tanks on December 13th from 07:00 to 07:20h local time with the following figures disclosed:

Draft Fore:	13,16	m	Draft Aft:	13,27	m	List:	0,00	Trim Correction Applied:	Yes	METRIC TONS		
Tank n°	Grade	Sounding/ Ullage (m)	Deg C	Total Vol Observed	Free Water		Gross Obs Vol (m3)	VCF TAB 54B	Density (S.G.) at 15° C	Gross Std Vol @15° C	IN VAC	IN AIR
HFO 1P	VLSFO	1,38 u	26,0	284,790	Nil	0,000	284,790	0,9920	0,9470	282,512	267,539	267,228
HFO 1S	VLSFO	4,58 u	26,0	4,200	Nil	0,000	4,200	0,9920	0,9470	4,166	3,946	3,941
HFO 2P	VLSFO	0,11 s	43,0	1,110	Nil	0,000	1,110	0,9797	0,9470	1,087	1,030	1,029
HFO 2S	VLSFO	0,14 s	27,0	1,390	Nil	0,000	1,390	0,9911	0,9256	1,378	1,275	1,274
HFO 3P	VLSFO	1,68 u	43,0	279,700	Nil	0,000	279,700	0,9796	0,9452	273,994	258,979	258,678
HFO 3S	VLSFO	1,65 u	43,0	283,000	Nil	0,000	283,000	0,9796	0,9452	277,227	262,035	261,730
HFO 1SERV	VLSFO	GAUGE	72,0	19,540	Nil	0,000	19,540	0,9571	0,9256	18,702	17,310	17,290
HFO 1 SETT	VLSFO	GAUGE	72,0	14,400	Nil	0,000	14,400	0,9571	0,9256	13,782	12,757	12,742
HFO 2 SERV	VLSFO	GAUGE	88,0	23,800	Nil	0,000	23,800	0,9449	0,9256	22,489	20,815	20,791
HFO 2SETT	VLSFO	GAUGE	88,0	26,000	Nil	0,000	26,000	0,9449	0,9256	24,567	22,740	22,713
OVERFLOW	VLSFO	4,78 u	21,0	4,780	Nil	0,000	4,780	0,9956	0,9312	4,759	4,432	4,426
Total							942,710			924,663	872,858	871,840

Sound: (U)=Ullage / (S)=Deep Sounding

Total quantity supplied as per the vessel's tanks:

Initial quantity disclosed.....: 355.620 mt
Final Quantity disclosed.....: 871,840 mt
Difference.....: 516,220 mt

5.2. Barge's tanks

The final sounding on the barge's tanks took place after the Barge's Inspector confirmed that the total amount ordered had been transferred from the barge tanks to the vessel's tanks from December 13th at 06:20 at 06:40 l/t being the following figures disclosed:

Barge's inspector figures.....: 2,134.388 mt
Surveyor's figures.....: 2,129.652 mt
Difference.....: - 3.576 mt
Flowmeter readings.....: OUT OF ORDER

AFTER RECEIVING								FLOWMETER	N/A m ³			
Draft Fore:	m		Draft Aft:	m		List:	0,00	Trim Correction Applied:	YES	METRIC TONS		
Tank nº	Grade	Sounding / Ullage	Deg C	Total Vol Observed	Free Water		Gross Obs Vol (m3)	VCF TAB 54B	Density (S.G.) at 15° C	Gross Std Vol @15° C	IN VAC	IN AIR
1 BB	VLSFO	4,602	37,0	81,220	Nil	0,000	81,220	0,9840	0,9452	79,920	75,541	75,453
1 BE	VLSFO	4,622	37,0	80,779	Nil	0,000	80,779	0,9840	0,9452	79,487	75,131	75,043
2 BB	VLSFO	1,708	44,5	405,000	Nil	0,000	405,000	0,9785	0,9452	396,293	374,576	374,140
2 BE	VLSFO	1,711	45,0	402,123	Nil	0,000	402,123	0,9782	0,9452	393,357	371,801	371,368
3 BB	VLSFO	4,426	44,0	164,157	Nil	0,000	164,157	0,9789	0,9452	160,693	151,887	151,711
3 BE	VLSFO	4,447	43,0	161,677	Nil	0,000	161,677	0,9796	0,9452	158,379	149,700	149,525
4 BB	VLSFO	1,730	44,0	403,816	Nil	0,000	403,816	0,9789	0,9452	395,295	373,633	373,198
4 BE	VLSFO	1,701	45,0	406,248	Nil	0,000	406,248	0,9782	0,9452	397,392	375,615	375,178
5 BB	VLSFO	4,639	38,0	97,966	Nil	0,000	97,966	0,9833	0,9452	96,330	91,051	90,945
5 BE	VLSFO	4,618	36,0	100,135	Nil	0,000	100,135	0,9847	0,9452	98,603	93,199	93,091
Total							2105,020	0,9789	0,9452	2060,816	1947,884	2129,652

6. Final figures

BDN.....: 549.001 mt (Barge's tanks)
Surveyor.....: 516,220 mt (Ship's tanks)
Difference.....: - 32,781 mt or -5,971%

Due to the huge difference between the BDN and the quantities ascertained through to the vessel's tanks the Master and the undersigned surveyor issued Letters of Protest against the suppliers.

Owing to the significant difference between BDN and vessel's figures a second sounding was carried out December 14 from 21:30 x 22:00h and the difference observed after bunkering was confirmed.

ROB after bunkering.....: 865.071 MT / Trim 0,11 m

7. Photographic report

7.1. Vessel's tanks – opening soundings





7.2. Sampling



8. Attachment

- Bunker Quantity Survey Report issued by undersigned surveyor
- Bunker Delivery Note (BDN) issued by Petrobras
- Barge Tank Measurement (BTM) issued by Petrobras
- Letter of Protest issued by vessel
- Letter of Protest issued by Naabsa

